IqraSec Academy | WAPT Phase 1

Task 1: Web Fundamentals

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Report title	Task 1: Web Fundamentals
Rooms / Labs covered	Web Application Basics; HTTP in Detail (TryHackMe)
Date completed	1 November 2026

1. Objective

The objective of this task was to understand how web applications are built and how they communicate over HTTP, so that vulnerabilities in that communication can be recognised in later tasks.

2. Concept / Understanding

Web application basics

- A web application has a front end (HTML, CSS and JavaScript that run in the browser) and a back end (server, application logic and database that run on the server).
- Because the front end runs on the user's machine, anyone can read or change it. Sensitive data such as passwords or API keys must never be left in HTML or JavaScript — this is sensitive data exposure.
- If a page shows user input without checking it, an attacker can inject their own HTML. This is HTML injection and is the basis of cross-site scripting.
- Components such as load balancers, CDNs, databases and web application firewalls (WAF) sit between the user and the application and affect how it behaves.

HTTP in detail

- HTTP is the protocol a browser uses to ask a server for resources. A request goes to the server and a response comes back. HTTPS is the same protocol encrypted with TLS.
- A request has a method (GET to fetch data, POST to send data, PUT to update, DELETE to remove), a path, headers and sometimes a body.
- A response has a status code, headers and a body. Common codes: 200 OK, 301/302 redirect, 400 bad request, 401 unauthorised, 403 forbidden, 404 not found, 500 server error.
- Headers carry extra information — Host, User-Agent, Cookie and Content-Type in requests; Set-Cookie, Content-Length and Server in responses.
- HTTP is stateless, so the server does not remember users between requests. Cookies solve this by storing a session identifier in the browser. If an attacker steals or guesses that identifier, they can take over the session.

3. Practical Work

- Created a TryHackMe account and completed the **Web Application Basics** room end-to-end, finishing all tasks in the room (100% room progress).
- Completed the **HTTP in Detail** room, including the tasks covering requests, responses, HTTP methods, status codes and headers (100% room progress).
- Read through the explanations of how a browser and a web server exchange requests and responses, and how headers/cookies carry state between them.

4. Tools Used

Tool	Purpose
TryHackMe	Completed the guided rooms and in-room questions
Web browser (Chrome)	Opened the rooms and read through the material

5. Evidence

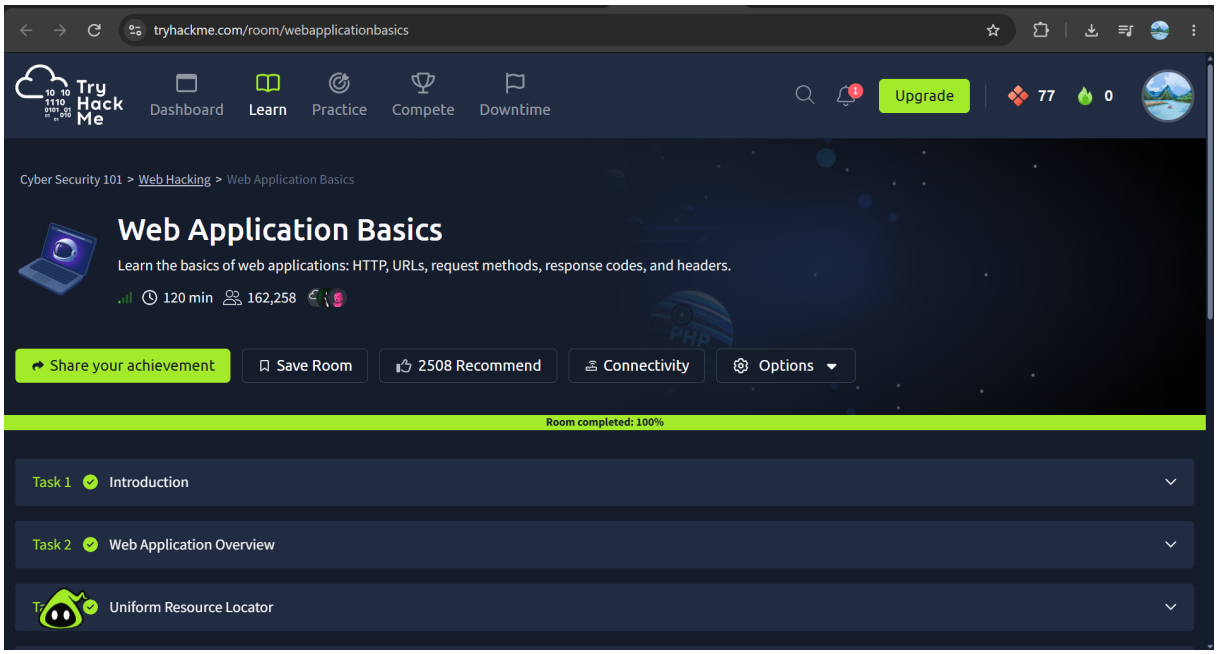


Figure 1: *Web Application Basics room completed at 100%.*

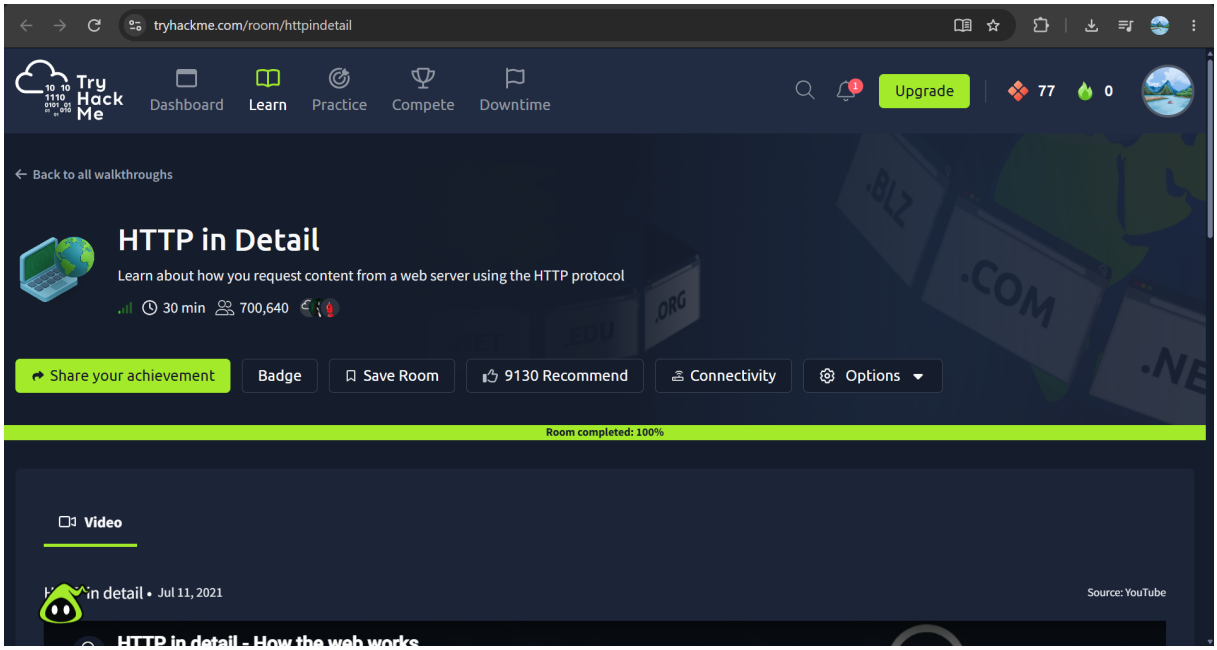


Figure 2: *HTTP in Detail room completed at 100%.*

■ Evidence gap: Only room-completion screenshots are available for this task. The submission guide asks for evidence of actual practical work, not just completion screens — e.g. an in-room answer being submitted, or a browser DevTools Network tab showing a real request's headers and cookies. Add one or two such screenshots before submitting if possible.

6. Result

Both rooms were completed and all in-room questions were answered. I can now describe how a request travels from the browser to the server and back, read the main parts of an HTTP request and response, and explain why cookies and headers matter for security.

7. Key Learnings

- Everything in the front end can be seen and changed by the user, so security checks must always happen on the server.
- HTTP status codes and headers give useful clues about how a server behaves, which helps later when testing an application.
- Cookies identify a user between requests, so weak or exposed session cookies can lead to account takeover.
- Unchecked user input can lead to HTML injection and, later, cross-site scripting (XSS).

Declaration: I confirm that this report documents my own work and that all screenshots were taken by me during the tasks described.

Name: Maryam Batool Date: 1 November 2026